Atom Rosales

Spatial data science · GIS · public health · operational analytics

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Profile

Spatial data science and public health professional specializing in GIS, remote sensing, UAS operations, surveillance analytics, and operational decision support. My work focuses on transforming environmental, field, treatment, and surveillance data into spatial systems that support vector-borne disease prevention, regulatory compliance, emergency response, and defensible public health operations.

My applied experience is rooted in mosquito control and integrated pest management, with transferable expertise in geospatial data infrastructure, scientific computing, UAS program management, remote sensing, and team leadership.

Core Competencies

- Spatial data science for public health surveillance, environmental monitoring, treatment planning, and vector-borne disease prevention
- GIS program administration, relational databases, dashboards, field data systems, and operational web applications
- R and Python workflows for reproducible reporting, surveillance summaries, treatment threshold analysis, QA/QC, and operational metrics
- UAS program leadership, operator training, aerial applications, and remote-sensing data acquisition
- Drone regulatory compliance, FAA Part 107 operations, Public COA management, operational waivers, and public-sector UAS risk management
- Team leadership, technical training, workflow development, and cross-functional coordination between field, laboratory, aviation, and operations staff

Education

Master of Science, Spatial Data Science

The Pennsylvania State University, State College, PA

May 2026

Post-Baccalaureate Certificate, Geographic Information Systems

The Pennsylvania State University, State College, PA

May 2024

Bachelor of Arts, Philosophy

University of North Texas, Denton, TX

December 2017

Honors Scholar

University of North Texas, Denton, TX

December 2017

Certifications and Licenses

- FAA Part 107 Remote Pilot Certificate
- Skydio X10 Certified Operator
- Skydio X10 Certified Instructor
- Florida Public Health Pest Control License
- Florida Aerial Applicator License
- Public Certificate of Authorization management for large-UAS spray operations
- FAA waiver experience, including beyond visual line of sight operations and operations over people

Selected Applied Research and Projects

Master's Capstone: Drone-Applied Granular Larvicide Swath Characterization

Developed a reproducible spatial analysis workflow to evaluate deposition patterns of VectoMax FG applied from a PrecisionVision 40X UAS. Used transect and grid-based collection data to calculate application rates, simulate overlapping swaths, assess deposition uniformity, and model spatial variation in delivered dosage.

- Processed field-collected granule deposition data from line-transect and grid-based experiments
- Calculated application rates, replicate variability, coefficient of variation, and swath-width performance metrics
- Simulated overlapping flight passes to evaluate candidate swath widths against a target application rate
- Fit spatial trend-surface models to estimate and visualize delivered dosage across the sampled grid
- Interpreted results for drone-based larvicide calibration, treatment consistency, regulatory compliance, and integrated vector management operations

Aerial Swath Analysis Platform

Developing a reproducible analytical and visualization program for processing aerial granular application data, evaluating swath width, modeling overlap, and supporting operational calibration of aerial application systems.

Drone-Based LiDAR Habitat Mapping

Applied UAS-based LiDAR and remote-sensing workflows to support characterization of mosquito production habitat, with emphasis on microtopography, field validation, and treatment prioritization.

Professional Experience

Director of Technical Development

Collier Mosquito Control District — Naples, Florida

March 2023–present

Lead spatial data science, GIS, UAS operations, and technical development initiatives that support public health mosquito-control operations. Translate surveillance, treatment, environmental, laboratory, and operational data into geospatial systems, analytical workflows, dashboards, and decision-support tools used across field operations, aviation, laboratory surveillance, and administration.

- Built and modernized geospatial decision-support systems that connect surveillance, treatment planning, field data collection, regulatory reporting, and operational review workflows
- Developed GIS tools, dashboards, web applications, and data pipelines that improve access to timely operational information for field, laboratory, aviation, and administrative teams
- Led spatial analysis projects evaluating vector species distribution, surveillance trends, treatment thresholds, operational performance, and environmental drivers of mosquito production
- Created reproducible analytical workflows using R, Python, GIS, dashboards, and automated reporting tools to improve consistency and expediency of operations
- Advanced the District's UAS program from individual mission support toward a managed operational capability, including operator coordination, training, documentation, safety practices, and equipment readiness
- Integrated drone-based photogrammetry, LiDAR, aerial imagery, and larvicide application workflows into mosquito surveillance, habitat assessment, treatment planning, and post-mission review
- Managed UAS regulatory compliance for FAA Part 107 operations, Public Certificate of Authorization requirements, and operational waivers
- Evaluated and implemented emerging technologies that improve surveillance, treatment operations, remote sensing, regulatory compliance, and public health decision-making

Technical Development Specialist

Collier Mosquito Control District — Naples, Florida

January 2023–March 2023

- Developed automated workflows using R, Jupyter Notebooks, Microsoft Power Automate, ArcGIS Pro script tools, and ArcGIS Online applications to reduce manual data handling and improve repeatability of operational reporting
- Built dashboards and web maps that made vector species distribution, seasonal abundance, arbovirus activity, and treatment-related data more accessible to operational staff

- Supported UAS photogrammetry, inspection, treatment, and image-processing workflows, helping translate drone-collected data into usable operational products
- Assisted with aerial and ground equipment calibration, swath analysis, and treatment-application data review to support more defensible application decisions
- Maintained drone hardware, firmware, payloads, and mission-readiness documentation to support reliable UAS operations
- Designed and 3D-printed custom operational components, including mosquito trap parts, nozzles, and drone-related components, to solve field and equipment constraints

Biologist

Collier Mosquito Control District — Naples, Florida

September 2022–December 2022

- Analyzed mosquito surveillance data in R to summarize abundance patterns, support treatment threshold calculations, and inform operational decision-making for vector species
- Cleaned, validated, and quality-checked trap-count data from remote surveillance tools, including BG-Counter traps, to improve reliability of surveillance summaries
- Developed ArcGIS Online dashboards for mosquito testing pools, mosquitofish distribution, treatment applications, and surveillance communication
- Supported laboratory disease-testing workflows by maintaining equipment readiness, reagent inventory, mosquito pool preparation, and related operational data records

Laboratory Technician

Collier Mosquito Control District — Naples, Florida

October 2022–January 2023

- Conducted RT-PCR testing on mosquito pools for pathogens including West Nile virus, Eastern equine encephalitis virus, and dengue virus
- Identified, sorted, and prepared mosquitoes for disease testing, surveillance reporting, and experimental use
- Set and retrieved mosquito surveillance traps, including BG-Sentinel traps, CDC light traps, and ovicups
- Supported the collection, preparation, and quality control of surveillance data used in public health mosquito-control decisions

Mosquito Surveillance Field Lead

The Woodlands Township Environmental Services — The Woodlands, Texas

March 2020–March 2022

- Coordinated adult trap assignments, larval surveillance activities, field routes, and seasonal staff operations to support consistent mosquito surveillance coverage
- Designed and implemented a mosquito abatement geodatabase using ArcGIS field applications, improving field data collection, route organization, and surveillance documentation
- Trained seasonal field technicians in adult trapping, larval surveillance, specimen handling, equipment use, and field data protocols
- Managed rearing workflows and quality control standards for field-collected mosquito specimens and biological control operations
- Translated field observations and surveillance data into operational priorities for mosquito-control planning

Administrative Assistant

Harris County Precinct 4 Biological Control Initiative — Houston, Texas

March 2019–March 2020

- Prepared insectary production metrics and summary reports used to track colony health, rearing output, and operational activity
- Maintained laboratory and office supply tracking to support continuity of biological control operations
- Assisted with rearing, maintenance, and documentation for biological control agents
- Supported coordination between administrative, laboratory, and field staff

Mosquito Surveillance Technician

The Woodlands Township Environmental Services — The Woodlands, Texas

May 2018–December 2018

- Set and retrieved CDC gravid traps, CDC light traps, GATs, and BG-Sentinel traps for mosquito surveillance
- Maintained and repaired surveillance and treatment equipment used in field operations
- Performed mosquito identification, pooling, and immunoassays on collected mosquitoes
- Assisted with public and community outreach events

Publications

Peer-reviewed Articles

Lucas, K. J., McDuffie, D., Gutierrez, I., Kacinsakas, S., **Rosales, A.**, Konieczny, O., & Luna, C. (2026). Susceptibility status of *Aedes aegypti* populations to naled in Collier County, Florida. *Journal of the American Mosquito Control Association*. Advance online publication. https://doi.org/10.2987/26-7290

Kuppe, C. R., Duett, M., Arber, K., **Rosales, A.**, Bonaccorsi, J., Oliva, D., Diclaro, J. W., II, & Qualls, W. A. (2026). Operational evaluation of unmanned aerial vehicle-applied granular larvicides in an integrated mosquito management program. *Acta Tropica*. https://doi.org/10.1016/j.actatropica.2026.108146

Martin, H., Reeves, L. E., Steele, G., **Rosales, A.**, Heinig, R., & Lucas, K. J. (2024). *Aedeomyia squamipennis*: A new genus and species record for Collier County, Florida, USA. *Journal of the American Mosquito Control Association, 40*(4), 174–177. https://doi.org/10.2987/24-7188

Reeves, L. E., Sloyer, K. E., Tyler-Julian, K., Heinig, R., **Rosales, A.**, Domingo, C., & Burkett-Cadena, N. D. (2023). *Culex* (*Phenacomyia*) *lactator* (Diptera: Culicidae) in southern Florida, USA: A new subgenus and species country record. *Journal of Medical Entomology, 60*(3), 478–486. https://doi.org/10.1093/jme/tjad023

Professional Articles and Trade Publications

Rosales, A. (2025, November 1). 3D Modeling and Manufacturing Techniques. *Wing Beats*.

Rosales, A. (2023, September 15). Improved Sample Tray for Adult Mosquito Identification. *Wing Beats*.

Rosales, A. (2023, September 15). State-Mandated Security Standards for Drone Use and Its Impact on Mosquito Control Operations. *Wing Beats*.

Presentations

Invited Talks

Rosales, A. (October 2025). Evaluation of Unmanned Aerial System Application of VectoBac FG+ for Control of *Culex nigripalpus* in a Continuously Flooded Habitat. Northwest Mosquito and Vector Control Association Annual Meeting, Island Park, ID.

Rosales, A. (March 2024). Taking Flight with Drone Photogrammetry: Operational Uses for Aerial Imagery. American Mosquito Control Association Annual Meeting, Dallas, Texas.

Rosales, A. (April 2024). Evolution of Unmanned Aerial Systems (UAS) at Collier Mosquito Control District. Florida Mosquito Control Association Annual Fly-In, Lehigh Acres, Florida.

Contributed Talks

Rosales, A. (March 2026). Evaluation of Unmanned Aerial System Application of VectoBac FG+ for Control of *Culex nigripalpus* in a Continuously Flooded Habitat. American Mosquito Control Association Annual Meeting, Portland, Oregon.

Rosales, A. (March 2026). Introducing the Aerial Swath Analysis Platform (ASAP): A Web Application for Modeling and Optimizing Granular Aerial Applications. American Mosquito Control Association Annual Meeting, Portland, Oregon.

Rosales, A. (November 2025). Evaluation of Unmanned Aerial System Application of VectoBac FG+ for Control of *Culex nigripalpus* in a Continuously Flooded Habitat. Florida Mosquito Control Association Annual Meeting, Port Charlotte, Florida.

Rosales, A. (March 2025). Evolution of Unmanned Aerial Systems (UAS) at Collier MCD. Florida Mosquito Control Association Annual Fly-In, Fort Myers, Florida.

Rosales, A. (March 2025). Drone-Based LiDAR Mapping of Mangrove Microtopography for Targeting *Aedes taeniorhynchus* Production Sites. American Mosquito Control Association Annual Meeting, San Juan, Puerto Rico.

Rosales, A. (November 2024). Where Have They Gone? Investigating the Missing *Culex nigripalpus* at Collier Mosquito Control District. Florida Mosquito Control Association Annual Meeting, Orlando, Florida.

Rosales, A. (April 2024). Overview of Unmanned Aerial System (UAS) Operations at Collier Mosquito Control District. Lee County Mosquito Control Aerial Workshop, Lehigh Acres, Florida.

Rosales, A. (January 2024). A Year in Review: Operational Assessment of AG-Mission at Collier Mosquito Control District. Florida Mosquito Control Association Annual Fly-In, Ellenton, Florida.

Rosales, A. (November 2023). Leveraging ArcGIS Online to Enhance Treatment Operations. Florida Mosquito Control Association Annual Meeting, Cape Coral, Florida.

Rosales, A. (April 2023). Winging It: Optimizing Aerial Treatment Operations with AG-Mission. Lee County Mosquito Control Aerial Workshop, Fort Myers, Florida.

Rosales, A. (March 2023). Evaluating BG-Counter Performance in Surveilling Collier County's Diverse Mosquito Populations. American Mosquito Control Association Annual Meeting, Reno, Nevada.

Rosales, A. (November 2022). Evaluating BG-Counter Performance in Surveilling Collier County's Diverse Mosquito Populations. Florida Mosquito Control Association Annual Meeting, Palm Coast, Florida.

Teaching and Workshops

Rosales, A. (January 2026). Integrated Pest Management. Advanced Mosquito Control, DODD Short Courses, Florida Mosquito Control Association.

Focuses on the history of integrated pest management in the US, surveillance-driven decision-making, establishing treatment thresholds, and leveraging freely accessible GIS data for enhancing integrated mosquito management practices.

Rosales, A. (January 2026). Mosquito Surveillance and Identification. Introduction to Mosquito Control, DODD Short Courses, Florida Mosquito Control Association.

Focuses on adult and larval surveillance, trap deployment and use cases, specimen handling, mosquito identification, data quality, and the public health role of surveillance.

Rosales, A., & Lefkow, N. (January 2024). Unmanned Aerial Systems (UAS) for Mosquito Control: Regulations, Operations, and Best Practices. DODD Short Courses, Florida Mosquito Control Association.

Rosales, A. (January 2022). Taking on Tables: Intermediate Excel for the Mosquito Control Professional. DODD Short Courses, Florida Mosquito Control Association.

Awards and Recognition

Cyrus R. Lesser Student Paper Competition, 1st Place Winner

Florida Mosquito Control Association, 2025

T. Wainwright Miller Scholarship

Florida Mosquito Control Association, 2025

James M. Robinson Memorial Award

Florida Mosquito Control Association, 2024

In appreciation for outstanding service to the Florida Mosquito Control Association through the development of an end-to-end digital workflow for the Collier Mosquito Control District's Mosquitofish program.

Industry Shadow Program and Stipend Awardee

American Mosquito Control Association, 2023

Tallahassee Days Stipend Awardee

Florida Mosquito Control Association, 2023

Professional Service and Leadership

- UAS Committee Chair, Florida Mosquito Control Association, 2022–2024
- A/V Committee Member, Florida Mosquito Control Association, 2023–2026
- DODD Committee Member, Florida Mosquito Control Association, 2024–2026
- Science and Technology Committee Member, American Mosquito Control Association